

Medical Insurance Cost Prediction using Linear Regression

Project Documentation

Introduction

This project predicts medical insurance charges using Linear Regression.

Problem Statement

Predict insurance charges using demographic and health-related features.

Objectives

Build, train and evaluate a Linear Regression model.

Dataset

Medical insurance dataset with features: age, sex, bmi, children, smoker, region and target: charges.

Preprocessing

Checked missing values, removed duplicates, encoded categorical variables, scaled features and split data into training and testing sets.

Model Evaluation

R^2 Score: 0.8069 (80.69%), MAE: 4177.05, MSE: 35478020.68, RMSE: 5956.34.

Conclusion

The model achieved good predictive performance and can be used as a baseline regression model.

Author

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Thank You

Thank you for reviewing this project.